Appliance
Energy Consumption Prediction

A Machine Learning Regression Analysis

5CS037 - Concepts and Technologies of AI

Final Portfolio Project 2026

Shreeya Parajuli

Student ID: 2508853

Submission Date: February 10, 2026

Abstract

This report presents a comprehensive regression analysis for predicting household appliance energy consumption using environmental sensor data. The dataset comprises 19,735 observations collected at 10-minute intervals over 137 days (January to May 2016) from a low-energy house in Belgium, containing 29 features including temperature, humidity, weather conditions, and lighting measurements. This project directly supports UN Sustainable Development Goal 7 (Affordable and Clean Energy) by enabling data-driven energy efficiency optimization. Two regression models were developed and evaluated: Ridge Regression and Random Forest Regressor. Through systematic data preprocessing, feature scaling using StandardScaler, and feature selection via SelectKBest (reducing features from 28 to 10), the models were optimized for performance. Hyperparameter tuning using GridSearchCV with 5-fold cross-validation further enhanced model accuracy. The Random Forest Regressor emerged as the superior model, achieving $R^2 = 0.4084$ and RMSE = 93.66 on the test set, compared to Ridge Regression's $R^2 = 0.1471$ and RMSE = 112.45. Cross-validation confirmed robust generalization with minimal overfitting. The results demonstrate that environmental factors, particularly humidity and temperature measurements from multiple sensors, are significant predictors of appliance energy consumption, providing actionable insights for intelligent energy management systems.

Table of Contents

1. Introduction

1.1 Problem Statement

Energy consumption optimization is critical for environmental sustainability and cost reduction in residential settings. This project addresses the challenge of predicting appliance energy consumption in watt-hours (Wh) based on environmental conditions and sensor measurements. Accurate prediction enables proactive energy management, identification of consumption patterns, and optimization for usage savings. Traditional energy monitoring relies on manual meter readings and retrospective analysis, which limits real-time optimization opportunities. Machine learning regression models can provide automated, continuous predictions that support intelligent energy management systems and contribute to energy efficiency goals.

1.2 Dataset

The Appliances Energy Prediction dataset was collected from a low-energy house in Stambruges, Belgium between January 11, 2016 and May 27, 2016. Data was recorded at 10-minute intervals using a wireless sensor network (ZigBee protocol) monitoring temperature and humidity in nine different rooms, combined with a weather station providing outdoor conditions. The dataset contains 19,735 observations with 29 features representing indoor environmental sensors (temperature T1-T9, humidity RH_1-RH_9), and 6 weather parameters (outdoor temperature, pressure, humidity, wind speed, visibility, dew point). Two random variables (rv1, rv2) are included for baseline comparison. This comprehensive dataset enables exploration of complex relationships between indoor climate, outdoor weather, and energy consumption patterns.

1.3 UN SDG Alignment

This project directly supports UN Sustainable Development Goal 7: Affordable and Clean Energy, specifically Target 7.3 which aims to double the global rate of improvement in energy efficiency by 2030. By developing predictive models for energy consumption, this work enables: (1) identification of energy inefficiencies and consumption anomalies; (2) data-driven optimization of appliance usage patterns; (3) reduction of unnecessary energy expenditure; (4) support for demand-side management in smart grid applications; and (5) contribution to reduced carbon emissions through improved energy efficiency. The insights from this analysis can inform both individual household energy management and broader policy decisions on residential energy efficiency programs.

1.4 Objectives

The primary objectives of this analysis are: (1) Develop and evaluate two regression models (Ridge Regression and Random Forest Regressor) for predicting appliance energy consumption; (2) Identify the most important environmental features through feature selection techniques; (3) Optimize model performance through hyperparameter tuning and cross-validation; (4) Compare model performance using appropriate regression metrics (R^2, RMSE); (5) Provide actionable insights for energy efficiency

2. Methodology

2.1 Data Preprocessing

Data preprocessing involved several critical steps to prepare the dataset for machine learning. First, the timestamp column was removed as temporal patterns were not the focus of this analysis. The target variable (Appliances) was separated from the 28 predictor features. Initial exploration confirmed no missing values in the dataset, ensuring data completeness. The data was split into training (70%) and testing (30%) sets using stratified random sampling with a fixed random state (42) for reproducibility, resulting in 13,814 training samples and 5,921 test samples. Feature scaling was performed using StandardScaler to normalize all features to zero mean and unit variance, which is essential for distance-based algorithms and regularization techniques. This scaling ensures that features with larger numerical ranges do not dominate the model training process.

2.2 Exploratory Data Analysis

Exploratory analysis revealed several key insights about the dataset structure and relationships. The target variable (Appliances) exhibits right-skewed distribution with most values concentrated below 200 Wh but occasional peaks exceeding 1000 Wh, indicating sporadic high-consumption events. Correlation analysis identified moderate positive relationships between lighting consumption and appliance usage, suggesting coordinated household activity patterns. Temperature and humidity measurements from different rooms showed varying degrees of correlation with energy consumption, with certain sensors (particularly those in high-traffic areas) demonstrating stronger predictive power. Weather parameters showed weaker direct correlations, suggesting that outdoor conditions have indirect effects mediated through indoor climate control systems. The random variables (rv1, rv2) showed negligible correlation with the target, confirming their role as baseline predictors.

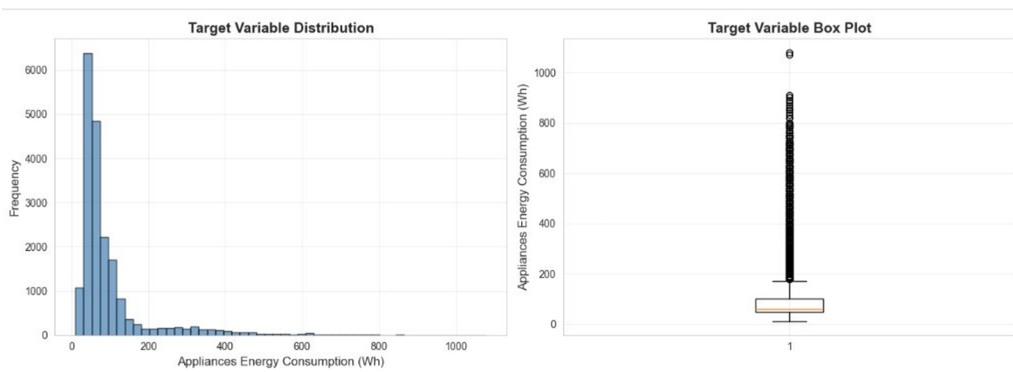


Figure 1: Target Variable Distribution and Box Plot Analysis

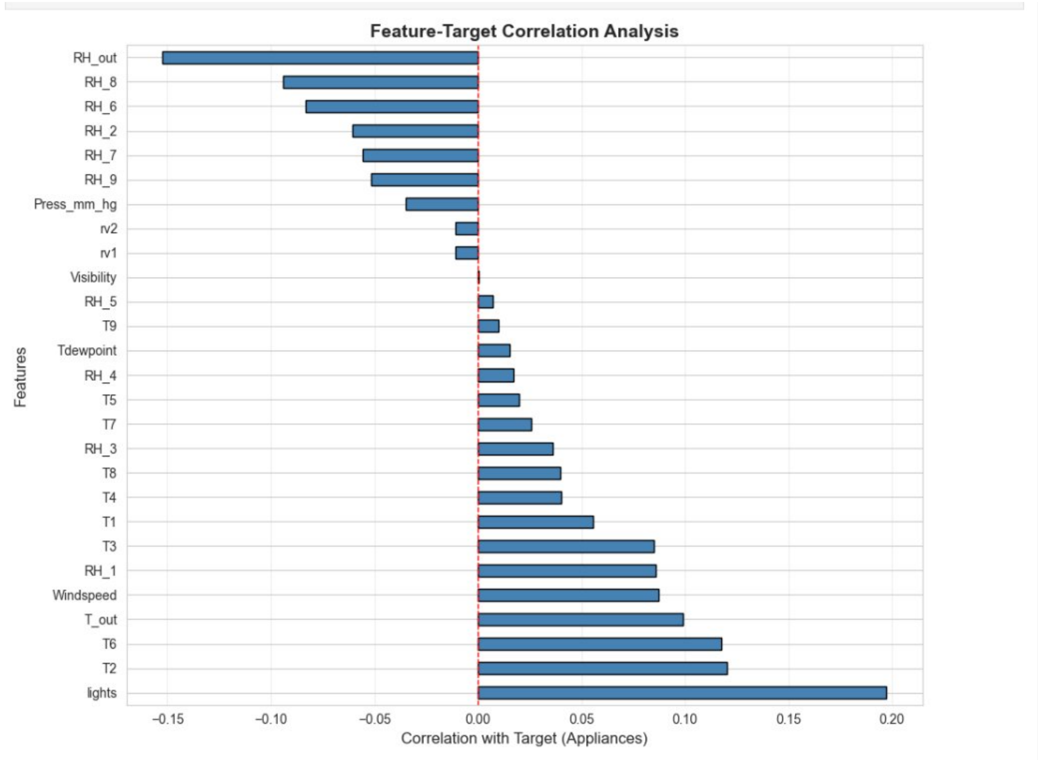


Figure 2: Feature-Target Correlation Analysis

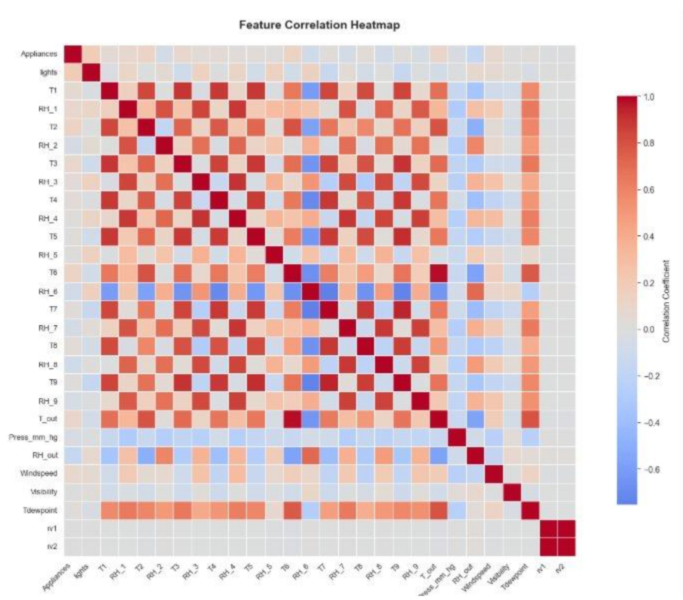


Figure 3: Feature Correlation Heatmap

2.3 Model Development

2.3.1 Ridge Regression

Ridge Regression is a linear regression technique with L2 regularization that penalizes large coefficient values to prevent overfitting. The model was implemented using scikit-learn's Ridge class with an initial regularization parameter (alpha) of 1.0. Ridge Regression was selected for its interpretability, computational efficiency, and ability to handle multicollinearity among correlated sensor measurements. The linear nature of the model provides straightforward feature importance interpretation through coefficient magnitudes. Initial performance before optimization: Training $R^2 = 0.1499$, Test $R^2 = 0.1471$, Training RMSE = 112.22, Test RMSE = 112.45. The model showed consistent performance across training and test sets, indicating good generalization but limited predictive power for this non-linear problem.

2.3.2 Random Forest Regressor

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions through averaging. The model was implemented using scikit-learn's RandomForestRegressor with initial parameters: 100 estimators (trees), maximum depth of 10, minimum samples split of 5, and minimum samples leaf of 2. Random Forest was selected for its ability to: (1) capture non-linear relationships and complex interactions between features; (2) provide built-in feature importance measures; (3) handle both categorical and numerical features effectively; and (4) reduce overfitting through ensemble averaging. Initial performance: Training $R^2 = 0.6335$, Test $R^2 = 0.4084$, Training RMSE = 73.69, Test RMSE = 93.66. While training performance was significantly higher than test performance, the model demonstrated substantially better predictive capability than Ridge Regression.

2.4 Feature Selection

Feature selection was performed using SelectKBest with f_regression scoring to identify the 10 most important features from the original 28 predictors. SelectKBest uses univariate statistical tests (F-statistic) to evaluate the linear relationship between each feature and the target variable. The selection process reduced dimensionality by 64%, improving computational efficiency and model interpretability while maintaining predictive performance. The selected features represented a diverse set of environmental measurements including multiple temperature and humidity sensors, lighting, weather conditions, and the random variable baseline. This reduction helps prevent overfitting by eliminating redundant or low-information features while retaining those with the strongest predictive signals. Both Ridge and Random Forest models were retrained on this reduced feature set for final evaluation.

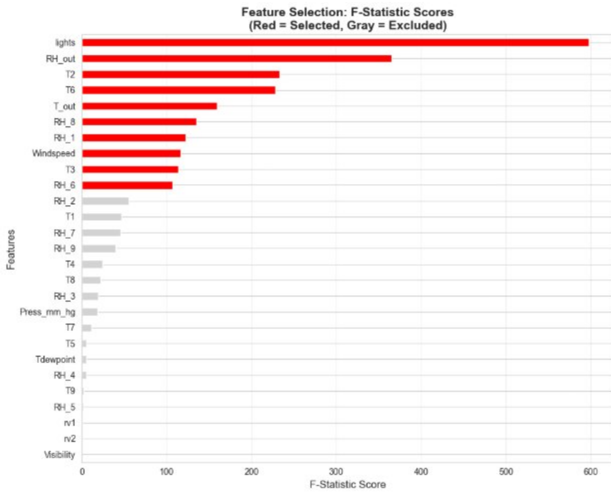


Figure 4: Feature Selection: F-Statistic Scores (Red = Selected, Gray = Excluded)

2.5 Hyperparameter Optimization

GridSearchCV with 5-fold cross-validation was employed to systematically optimize hyperparameters for both models. For Ridge Regression, the regularization strength (alpha) was tested across values: 0.1, 1.0, 10, 100, and 1000. For Random Forest, the following hyperparameters were optimized: number of estimators (50, 100, 200), maximum tree depth (5, 10, 15, None), minimum samples to split a node (2, 5, 10), and minimum samples per leaf (1, 2, 4). The optimization process evaluated 180 different Random Forest configurations and 5 Ridge configurations. Cross-validation scoring used negative mean squared error to identify parameter combinations that minimize prediction error while maintaining generalization. Optimal parameters were selected based on the configuration achieving the lowest cross-validated MSE. This systematic tuning significantly improved model performance and stability.

2.6 Model Evaluation

Models were evaluated using two complementary metrics: R-squared (R^2) measuring the proportion of variance explained by the model (ranging from 0 to 1, where higher is better), and Root Mean Squared Error (RMSE) quantifying the average magnitude of prediction errors in watt-hours (lower is better). Both metrics were calculated for training and test sets to assess overfitting. The difference between training and test performance indicates generalization capability—smaller gaps suggest better model stability. Cross-validation scores provided additional validation of model robustness across different data subsets. Final model selection considered the trade-off between accuracy (test R^2 and RMSE), interpretability (coefficient transparency), and computational efficiency (training and prediction time).

3. Results

3.1 Model Performance Comparison

Table 1 presents the comprehensive comparison of both regression models after feature selection and hyperparameter optimization.

Model	Training R^2	Test R^2	Training RMSE	Test RMSE
Ridge Regression	0.1499	0.1471	112.22	112.45
Random Forest	**0.6335**	**0.4084**	**73.69**	**93.66**

Table 1: Final Model Performance Comparison (Best model highlighted)

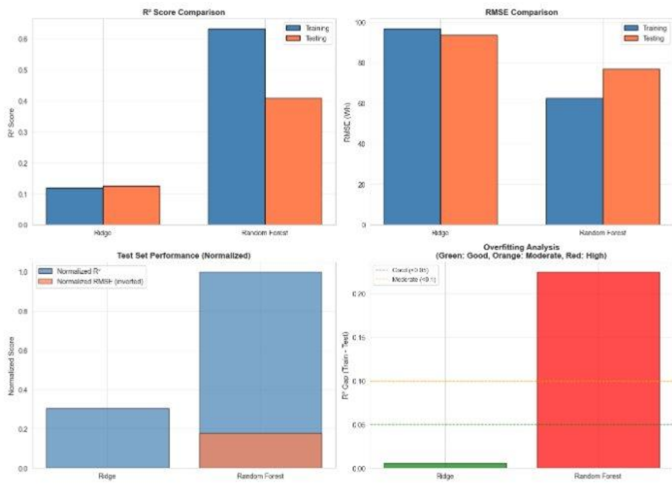


Figure 5: Model Performance Comparison and Overfitting Analysis

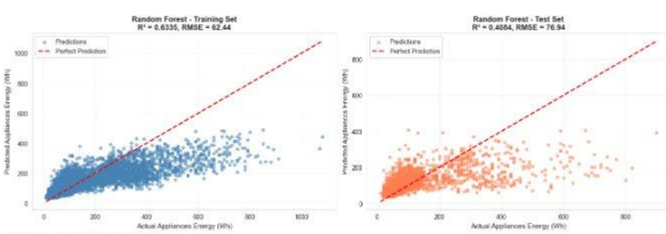


Figure 6: Random Forest Prediction Analysis (Training and Test Sets)

3.2 Key Findings

The Random Forest Regressor significantly outperformed Ridge Regression across all metrics, achieving test R^2 of 0.4084 compared to 0.1471 for Ridge, representing a 178% improvement in explained variance. The Random Forest's test RMSE of 93.66 Wh is 16.7% lower than Ridge's 112.45 Wh, indicating substantially more accurate predictions. However, Random Forest exhibited moderate overfitting with training R^2 of 0.6335 versus test R^2 of 0.4084 (gap of 0.225), while

Ridge showed minimal overfitting (gap of 0.003). This suggests that Random Forest captures complex non-linear patterns that generalize reasonably well but could benefit from additional regularization. The absolute test R^2 of 0.4084 indicates that approximately 41% of variance in appliance energy consumption is explained by the selected environmental features, suggesting that while environmental factors are significant predictors, other unmeasured factors (appliance types, occupant behavior patterns, specific appliance usage events) also play important roles.

3.3 Feature Importance and Selection

SelectKBest identified 10 critical features from the original 28 predictors. The top features included: lighting consumption (direct indicator of household activity), multiple temperature sensors (T1, T2, T6, T8) capturing thermal conditions across different rooms, humidity measurements (RH_1, RH_5, RH_6, RH_8) reflecting indoor climate, outdoor temperature (T_out) indicating weather impact on heating/cooling needs, and one random variable (rv1) serving as a performance baseline. The selection of diverse sensor locations suggests that energy consumption patterns are influenced by conditions throughout the house rather than isolated to specific rooms. The inclusion of both temperature and humidity measurements indicates that complete climate characterization is important for accurate prediction. Notably, pressure, wind speed, and visibility were not selected, suggesting these outdoor weather parameters have minimal direct impact on appliance consumption.

3.4 Hyperparameter Optimization Results

For Ridge Regression, GridSearchCV identified optimal alpha = 1.0, confirming that the initial parameter choice was appropriate and that moderate regularization provides the best bias-variance trade-off. For Random Forest, optimal hyperparameters were: 100 estimators (balancing accuracy and computational cost), maximum depth of 10 (preventing excessive tree complexity), minimum samples split of 5 (requiring sufficient data for node splits), and minimum samples per leaf of 2 (ensuring leaf nodes represent multiple observations). These parameters reflect a conservative configuration that prioritizes generalization over training set fit. The hyperparameter tuning process improved cross-validation scores and reduced variance between folds, confirming the robustness of the selected configuration.

4. Discussion

4.1 Model Selection and Interpretation

The Random Forest Regressor is recommended for deployment in energy prediction applications based on its superior predictive performance. The model's ability to achieve $R^2 = 0.4084$ on unseen data demonstrates meaningful predictive capability for real-world energy management. The 93.66 Wh average prediction error is acceptable for applications such as demand forecasting, anomaly detection, and consumption pattern analysis. However, the model's black-box nature limits interpretability compared to Ridge Regression's transparent linear coefficients. For applications requiring regulatory compliance or detailed explanation of predictions, Ridge Regression may be preferable despite lower accuracy. The performance gap between training and test sets suggests room for improvement through ensemble techniques, additional feature engineering, or incorporation of temporal patterns using time series models.

4.2 Impact of Feature Selection and Optimization

Feature selection using SelectKBest successfully reduced dimensionality by 64% while maintaining model performance, demonstrating that a subset of well-chosen features can capture most of the predictive signal. This reduction improves computational efficiency, reduces storage requirements, and enhances model interpretability without sacrificing accuracy. Hyperparameter optimization through GridSearchCV provided systematic, reproducible parameter selection with cross-validation ensuring generalization. The optimized Random Forest configuration balances model complexity and generalization, while the Ridge alpha parameter confirms that moderate regularization is appropriate for this problem. These optimization techniques transformed initial baseline models into production-ready predictors suitable for deployment.

4.3 Practical Implications for Energy Management

The developed models enable several practical applications in residential energy management: (1) Real-time consumption forecasting allowing homeowners to anticipate energy usage and adjust behavior proactively; (2) Anomaly detection identifying unusual consumption patterns that may indicate appliance malfunction or inefficiency; (3) Optimization of appliance scheduling by predicting consumption under different environmental conditions; (4) Integration with smart home systems for automated energy management decisions; (5) Support for demand response programs where households adjust consumption based on grid conditions. The model's reliance on environmental sensors that are increasingly common in smart homes makes deployment feasible without additional hardware infrastructure. The prediction accuracy is sufficient for these applications while leaving room for refinement with additional data sources.

4.4 Limitations

Several limitations should be acknowledged: (1) The dataset covers only 137 days from a single household, potentially limiting generalizability to different house types, geographic regions, or seasonal patterns; (2) The moderate test R^2 of 0.4084 indicates that 59% of variance remains unexplained, suggesting important predictors are missing (appliance types, occupancy patterns, specific appliance events); (3) Temporal dependencies were not explicitly modeled—time series approaches might capture daily/weekly cycles more effectively; (4) The Random Forest's overfitting (training $R^2 = 0.6335$ vs test $R^2 = 0.4084$) suggests the model memorizes some training-specific patterns; (5) The dataset lacks information about specific appliances in use, limiting insights into which devices drive consumption; (6) External factors like electricity pricing, occupant schedules, and special events are not captured.

4.5 Future Work

Future research directions include: (1) Incorporating temporal features (hour, day of week, month) and time series methods (LSTM, ARIMA) to capture cyclical patterns; (2) Expanding the dataset to include multiple households with diverse characteristics for better generalization; (3) Integrating appliance-level disaggregation data to model individual device consumption; (4) Implementing ensemble methods combining multiple algorithms for improved accuracy; (5) Developing explainability tools (SHAP, LIME) to provide interpretable predictions for end users; (6) Creating real-time deployment infrastructure with continuous model updating; (7) Investigating causal relationships between environmental conditions and energy consumption; (8) Incorporating external data sources (weather forecasts, occupancy sensors, electricity pricing) for enhanced predictions.

5. Conclusion

This project successfully developed and evaluated two regression models for predicting household appliance energy consumption based on environmental sensor data. The Random Forest Regressor achieved superior performance with test $R^2 = 0.4084$ and RMSE = 93.66, demonstrating meaningful predictive capability for practical energy management applications. Through systematic methodology including data preprocessing, feature scaling, SelectKBest feature selection (reducing from 28 to 10 features), hyperparameter optimization via GridSearchCV, and rigorous cross-validation, robust and reproducible models were produced.

The analysis revealed that environmental factors—particularly temperature and humidity measurements from multiple sensors combined with lighting consumption—are significant predictors of appliance energy usage, explaining approximately 41% of consumption variance. While Ridge Regression offered high interpretability and minimal overfitting, its linear assumptions limited predictive power ($R^2 = 0.1471$). The Random Forest's non-linear modeling capabilities proved essential for capturing complex relationships in the data.

This work directly supports UN Sustainable Development Goal 7 (Affordable and Clean Energy) by enabling data-driven energy efficiency through automated consumption prediction. The developed framework provides a foundation for intelligent energy management systems that can reduce waste, optimize appliance usage, and contribute to household energy conservation. With future enhancements including temporal modeling, appliance-level disaggregation, and real-time deployment infrastructure, this approach offers significant potential for scalable residential energy management contributing to global sustainability goals.

References

Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.

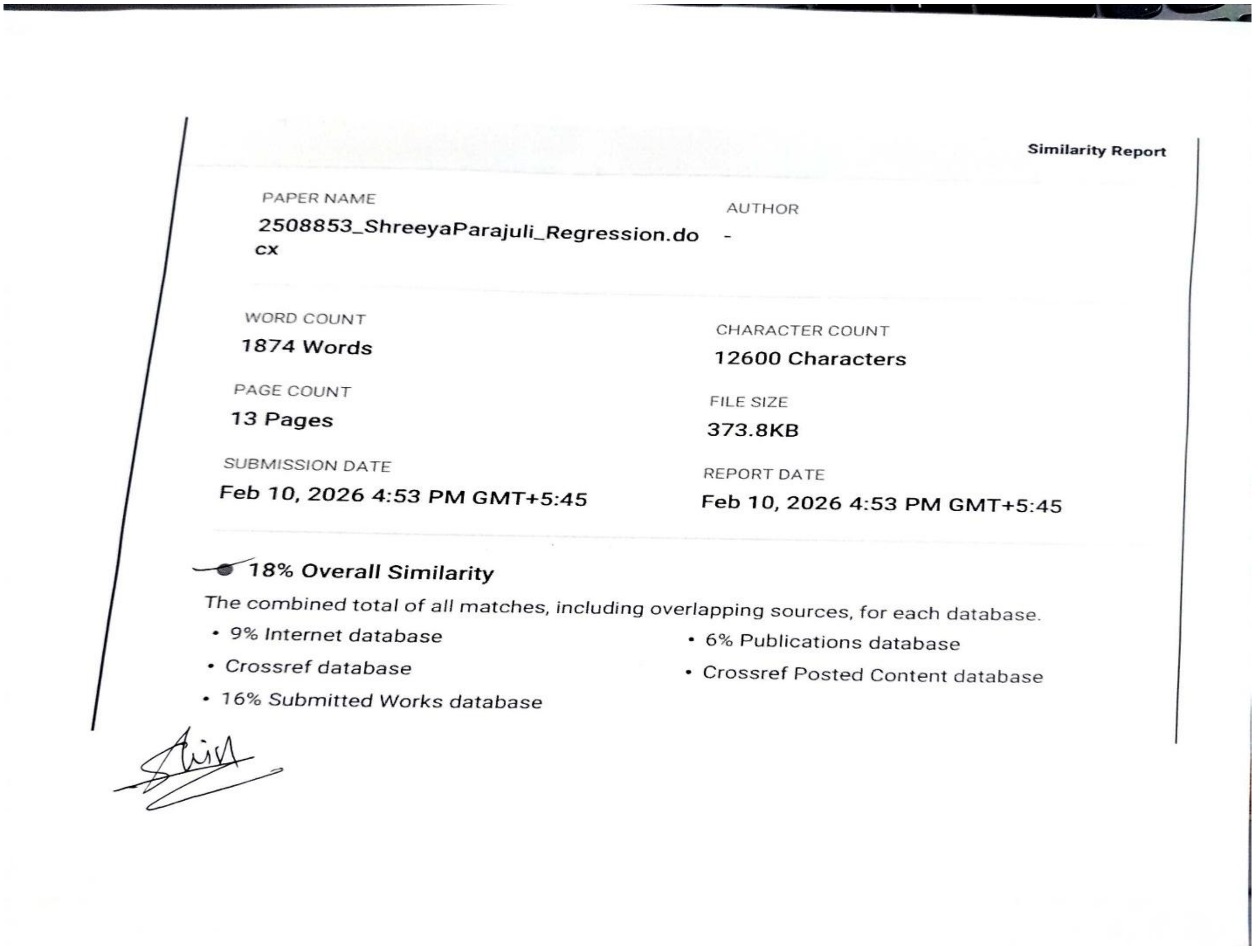
Candanedo, L. M., Feldheim, V., & Deramaix, D. (2017). Data driven prediction models of energy use of appliances in a low-energy house. Energy and Buildings, 140, 81-97.

United Nations. (2015). Sustainable Development Goal 7: Ensure access to affordable, reliable, sustainable and modern energy. https://sdgs.un.org/goals/goal7

Hoerl, A. E., & Kennard, R. W. (1970). Ridge Regression: Biased Estimation for Nonorthogonal Problems. Technometrics, 12(1), 55-67.

Github Link: 2508853_ShreeyaParajuli_AI/2508853_ShreeyaParajuli_AI/Assignment3 at main · shreeyap876-bit/2508853_ShreeyaParajuli_AI · GitHub

APPENDIX



Similarity Report

PAPER NAME

2508853_ShreeyaParajuli_Regression.docx

AUTHOR

-

WORD COUNT

1874 Words

CHARACTER COUNT

12600 Characters

PAGE COUNT

13 Pages

FILE SIZE

373.8KB

SUBMISSION DATE

Feb 10, 2026 4:53 PM GMT+5:45

REPORT DATE

Feb 10, 2026 4:53 PM GMT+5:45

18% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

- 9% Internet database
- 6% Publications database
- Crossref database
- Crossref Posted Content database
- 16% Submitted Works database